lec51

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1 Multi-class Classification

Recall:

- Given $(X, Y)_{i=1}^{n}$, classify new point $y_i \in \mathcal{L}$ given x_i, where $\mathcal{L}$ is the class of labels
- Since labels are discrete, evaluate with confusion matrix rather than R^2
- Accuracy:
$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$
- Typically optimize multi-classification models with cross-entropy loss

Confusion Matrix

	Predicted a	Predicted b	Predicted c
Actual a	n_{aa}	n_{ab}	n_{ac}
Actual b	n_{ba}	n_{bb}	n_{bc}
Actual c	n_{ca}	n_{cb}	n_{cc}

- **Precision**:
$$\text{Precision}_{\ell \in \mathcal{L}} = \frac{\text{\# of True Positives}}{\text{\# of Predicted Positives}} = \frac{TP}{TP + FP}$$
- Ex. Precision for class c is $\frac{n_{cc}}{n_{ac}+n_{bc}+n_{cc}}$
- **Recall**:
$$\text{Recall}_{\ell \in \mathcal{L}} = \frac{\text{\# of True Positives}}{\text{\# of Actual Positives}} = \frac{TP}{TP + FN}$$
- Ex. Recall for class c is $\frac{n_{cc}}{n_{ca}+n_{cb}+n_{cc}}$

Remarks:

- Precision and recall are used to mitigate class imbalance.

- Precision asks for the accuracy rate of positive classifications, so it is used when false positives are costly (e.g. spam email filter flagging an email as spam, when it isn't)
- Recall asks for the detection rate of positive cases. It is used when false negatives are costly (e.g. cancer diagnosis)

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import precision_score, classification_report
X_train, y_train, X_test, y_test = train_test_split(X, y)
model = LogisticRegression()
model.fit(X_train, Y_train)
y_pred = model.predict(X_test)
precision_score(y_test, y_pred)
clasification_report(y_test, y_pred) # gives precision/recall for minority and majority
```